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of ENERGY

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Science

# Rubin 2036: Technical Summary of Previous Workshops & Whitepapers

**Aaron Roodman**

Rubin 2036 Workshop, Aug 3 2026



# Previous Whitepapers

## REPORT ON LSST NEXT-GENERATION INSTRUMENTATION WORKSHOP, APRIL 11, 12 2019

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## Astro2020 APC White Paper Next Generation LSST Science

Type of Activity: ☒ Ground Based Project ☐ Space Based Project  
☐ Infrastructure Activity ☐ Technological Development Activity  
☐ State of the Profession Consideration ☐ Other

☐ Planetary Systems ☒ Star and Planet Formation  
☐ Compact Objects ☒ Cosmology and Fundamental Physics  
☐ Resolved Stellar Populations and their Environments  
☐ Multi-Messenger Astronomy and Astrophysics

## Astro2020: Activity, Project, and Statement of the Profession Consideration White Paper

## Future Use of the LSST Facility: In Team

## Astro2020 Science White Paper

## LSST Narrowband Filters

Thematic Areas: ☐ Planetary Systems ☐ Star and Planet Formation  
☐ Formation and Evolution of Compact Objects ☒ Cosmology and Fundamental Physics  
☒ Stars and Stellar Evolution ☒ Resolved Stellar Populations and their Environments  
☐ Multi-Messenger Astronomy and Astrophysics

☒ Planetary Systems ☒ Star and Planet Formation  
☒ Compact Objects ☒ Cosmology and Fundamental Physics  
☒ Resolved Stellar Populations and their Environments  
☒ Multi-Messenger Astronomy and Astrophysics

## Snowmass2021 Cosmic Frontier White Paper: Rubin Observatory after LSST

Bob Blum<sup>1</sup>, Seth W. Digel<sup>2,3</sup>, Alex Drlica-Wagner<sup>4,5</sup>, Salman Habib<sup>6,7</sup>, Katrin Heitmann<sup>7</sup>,  
Mustapha Ishak<sup>8</sup>, Saurabh W. Jha<sup>9</sup>, Steven M. Kahn<sup>2,3,10</sup>, Rachel Mandelbaum<sup>11</sup>, Phil Marshall<sup>2,3</sup>,  
Jeffrey A. Newman<sup>12</sup>, Aaron Roodman<sup>2,3</sup>, and Christopher W. Stubbs<sup>13</sup>

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# Workshops & Whitepapers

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- Cosmic Visions Dark Energy
  - Panel(s) met from 2015-1028 to examine broad questions about future Dark Energy projects
  - Multiple workshops: UChicago 2016, [LBL 2017](#)
  - Reports: [\[1604.07626\] Cosmic Visions Dark Energy: Science](#) [\[1604.07821\] Cosmic Visions Dark Energy: Technology](#) [Small Projects Portfolio](#)
- LSST Next Generation Workshop 2019
  - [\[1905.04669\] Report on LSST Next-generation Instrumentation Workshop, April 11, 12 2019.](#)
- Astro2020 Decadal Survey
  - White-papers: [\[1907.08945\] Next Generation LSST Science](#) [\[1907.10487\] Future Uses of the LSST Facility: Input from the LSST Project Science Team](#)
- 2022 Snowmass & P5
  - White-papers: [\[2203.07220\] Snowmass2021 Cosmic Frontier White Paper: Rubin Observatory after LSST](#) [\[2209.08654\] Report of the Topical Group on Dark Energy and Cosmic Acceleration: Complementarity of Probes and New Facilities for Snowmass 2021](#)
  - Report: [\[2209.08049\] Snowmass2021 Cosmic Frontier: Report of the CF04 Topical Group on Dark Energy and Cosmic Acceleration in the Modern Universe](#)

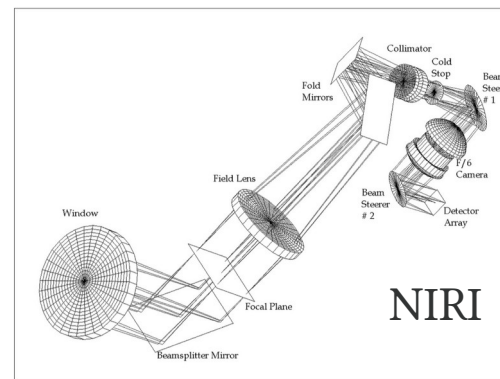
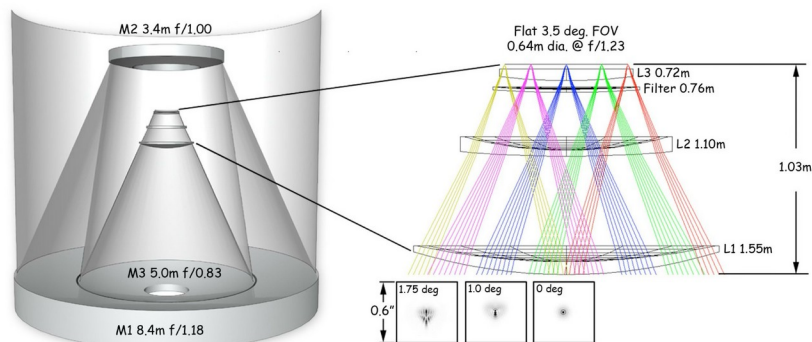
# Review of Future Technical Directions for Rubin

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- Replace Si CCD Focal Plane with IR image sensors
- Replace Si CCD Focal Plane with single-photon wavelength measuring sensors such as MKIDS
- Replace Imager with Multi-Object Spectrograph, Fiber fed (or Grism)
- Addition of Narrow and Medium Band Filters

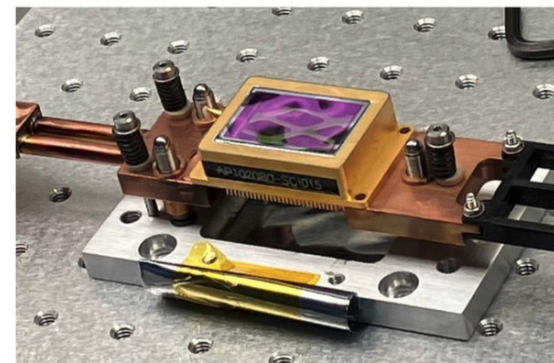
# IR Sensor Focal Plane

- Extend survey coverage into the IR?
- State of the art HxRG HgCdTe sensors are of order 1M\$ each
- Lack of a cold stop in the LSST Optical design limits wavelengths to less than 1.8 micron
  - For longer wavelengths, a combination of a Cold stop, cooled Focal Plane window and Cryostat are needed, to prevent thermal IR loading
  - Rubin's optical design has a *prime-focus* configuration with no collimation and no stops



# IR Sensor Focal Plane (cont)

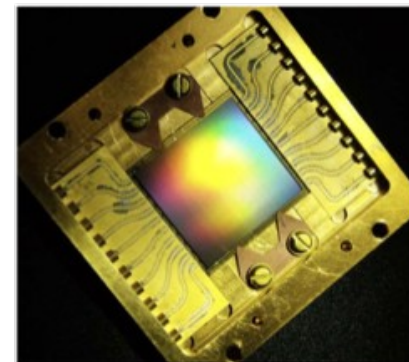
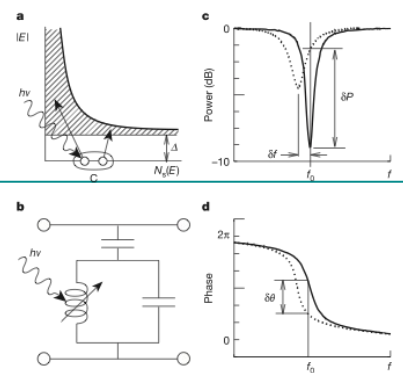
- Y, J, H band with InGaAs (to 1.7 micron) or Ge CCD
- See [WINTER, 1sq.deg. InGaAs camera](#) on Palomar 1m
  - IR time-domain wide field instrument
  - AP1020 sensors, 1920x1080 with 15 micron pixels
  - Designed for -40C operation, with 120-370 e-/sec/pix dark current and 50-70 e- RN
  - QE between 6-10%
  - J-band depth of 18.5
- Decadal survey paper comments:
  - LSST Camera lens have AR coating designed for 1100nm, would require recoating
  - Guess-timated ~100M\$
- Is there a path to an InGaAs NIR Focal Plane?
  - Evaluation of possible depth with optimistic sensor development
  - Key parameters: Temperature, RN, Dark Current, QE, Fill Factor
  - Evaluate further sensor development and costs



WINTER InGaAs sensor

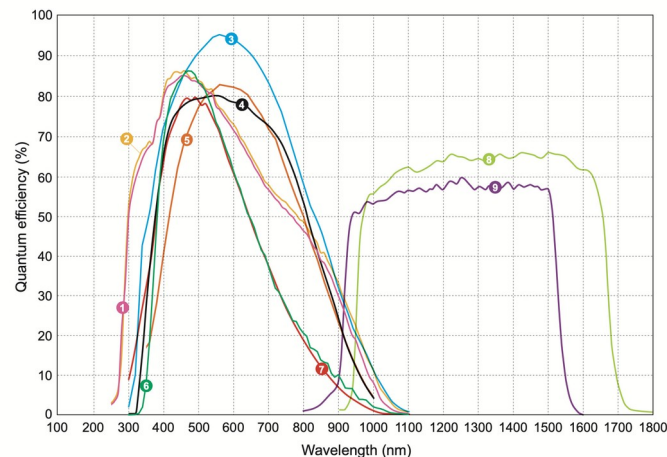
# MKIDS Focal Plane

- Kinetic Inductance detector
- Ultimate Photon Detector:
  - Wide-field Mosaic with small Plate Scale
  - Measure wavelength of every individual photon
  - Record photon time to 1 millisecond
- MKIDS have promise of measuring both wavelength and time, but
  - Talk in 2019 workshop pointed to current Energy resolution of  $R \sim 10$  projecting to  $R \sim 50$
  - See [Zobrist et.al 2022](#) which reached  $R \sim 20$
  - MEC at SUBARU has 20k 150 micron pixels, operating at 90mK, with  $R \sim 5-7$
  - XKID at Magellan
  - Both MEC and XKID are used with AO systems
- See comment above about Cold Stops...



# Silicon CMOS and Single Photon sensors

- Fast, very low-noise, CMOS sensors or Cameras are now commercially available.
- Hamamatsu qCMOS
  - RN less than  $1e^-$  at 120 frames/sec
  - Thin Silicon, so NIR QE is poor
  - Full well of  $7ke^-$
- Teledyne CIS
  - RN of  $2e^-$  at 8 frames/sec
  - (10 micron thick Silicon)
  - 3-side buttable
- Lincoln Labs SiSeRO & LBL/FNAL Skipper CCDs
  - Aiming for single-photon sensitivity
  - Effective read-out rate is optimized for spectroscopy



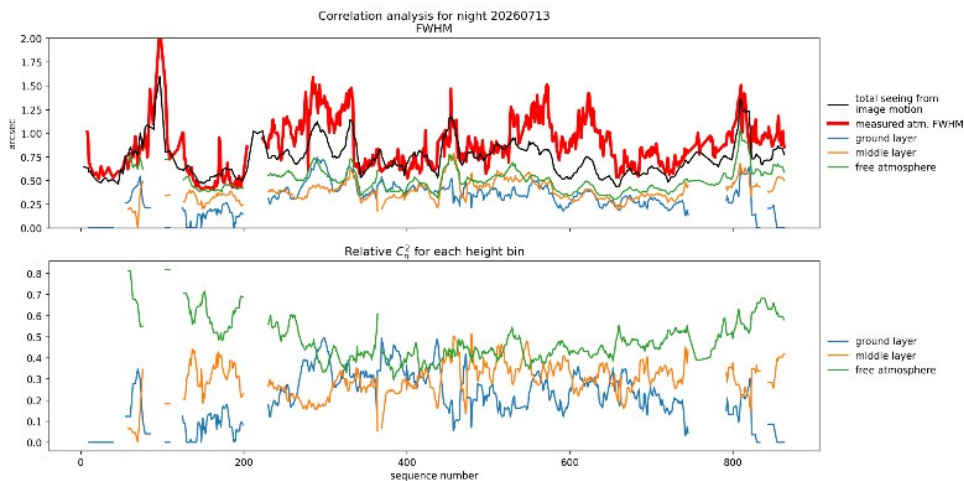
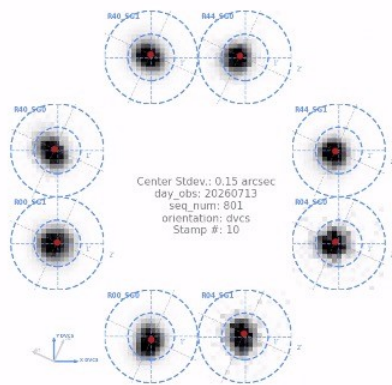
# Conversion to a Multi-Object Spectrograph

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- Take advantage of Rubin's large aperture and convert to a fiber-fed MOS
- Rubin's optical system presents challenges:
  - Fast beam with F# 1.23 is a poor match to typical fiber NA (of 0.2)
    - We have an LDRD project underway to evaluate the NA of conical fiber tips to improve throughput at F#1.2, as a technology demonstrator
    - Move to high NA fiber
  - Rubin's optical system is somewhat non-telecentric (6 deg at the edge of the FoV), so fibers would have to match that to optimize throughput
  - Relatively short focal length allows less room for ~5-10k fiber positioners and would require smaller patrol radius and smaller diameter fibers
  - Atmospheric Dispersion Corrector would be required
- Spectroscopy would yield Rubin's most distinctive feature: fast survey capability

# Ground-Layer AO

- Rubin's secondary mirror is nearly conjugate to the primary, so a tip/tilt secondary could correct common-mode turbulence
  - Correct with a Fast Active secondary mirror
    - M2 is 3.4m and 2700kg, so would require a lighter-weight cell and a counterweight
  - Use the LSSTCam Guider at 9Hz to isolate ground layer from common mode Image motion
    - Rubin study using the Guider to measure tomographic seeing, [Jackie Zhang at RCW](#)
    - Near-term goal to quantify possible gains from both Guiding and a GLAO



# Continued LSST-style Surveying

- How do various measurements scale with Time?
- See 2019 Decadal survey paper:

Table 1: Various science metrics as functions of survey duration.

| Quantity                             | Year 1 | Y3   | Y5   | Y8   | Year 10 | Y12  |
|--------------------------------------|--------|------|------|------|---------|------|
| $r_5$ coadd <sup>a</sup>             | 26.3   | 26.8 | 27.1 | 27.4 | 27.5    | 27.6 |
| $\sigma(i=25)^b$                     | 0.12   | 0.07 | 0.06 | 0.05 | 0.04    | 0.04 |
| color vol. <sup>c</sup>              | 316    | 20   | 6    | 1.7  | 1       | 0.6  |
| # of visits <sup>d</sup>             | 83     | 248  | 412  | 660  | 825     | 990  |
| $\sigma_\pi$ ( $r=24$ ) <sup>e</sup> | 9.5    | 5.5  | 4.2  | 3.3  | 3.0     | 2.7  |
| $\sigma_\mu$ ( $r=24$ ) <sup>f</sup> | 32     | 6.1  | 2.8  | 1.4  | 1.0     | 0.8  |

<sup>a</sup> The coadded depth in the  $r$  band (AB,  $5\sigma$ ; point sources).

<sup>b</sup> The photometric error for a point source with  $i = 25$ .

<sup>c</sup> The volume of the 5-dimensional color space, normalized by the final value.

<sup>d</sup> The number of visits per sky position (summed over all bands).

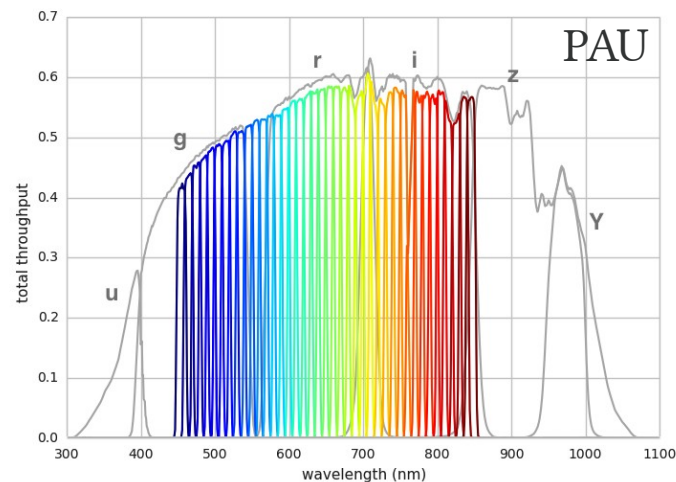
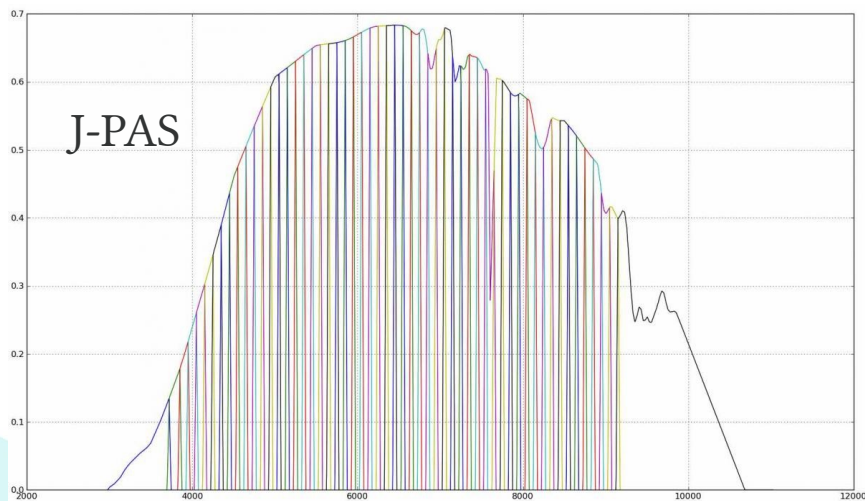
<sup>e</sup> The trigonometric parallax accuracy for a point source with  $r=24$  (milliarcsec).

<sup>f</sup> The proper motion accuracy for a point source with  $r=24$  (milliarcsec/yr).

- Also concludes that DETF Figure of Merit would continue to improve as  $1/\text{Time}$ , as would completeness of Hazardous Asteroid detection

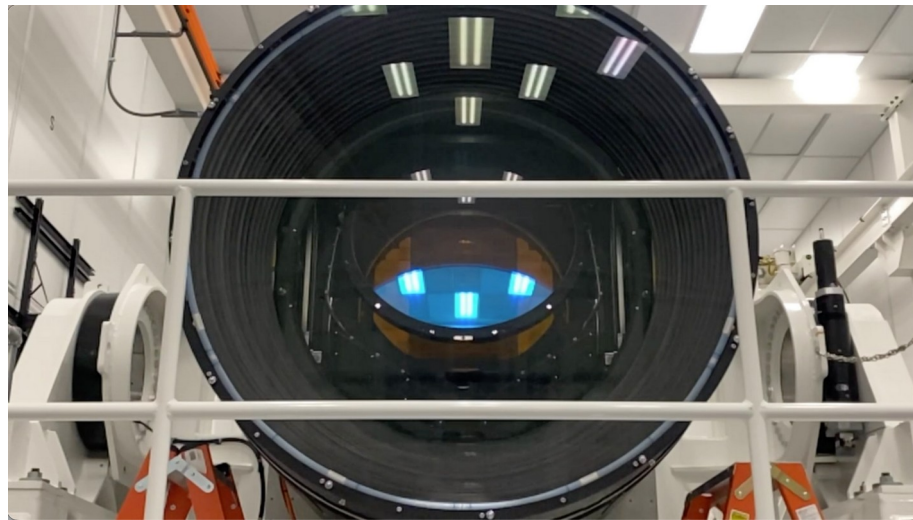
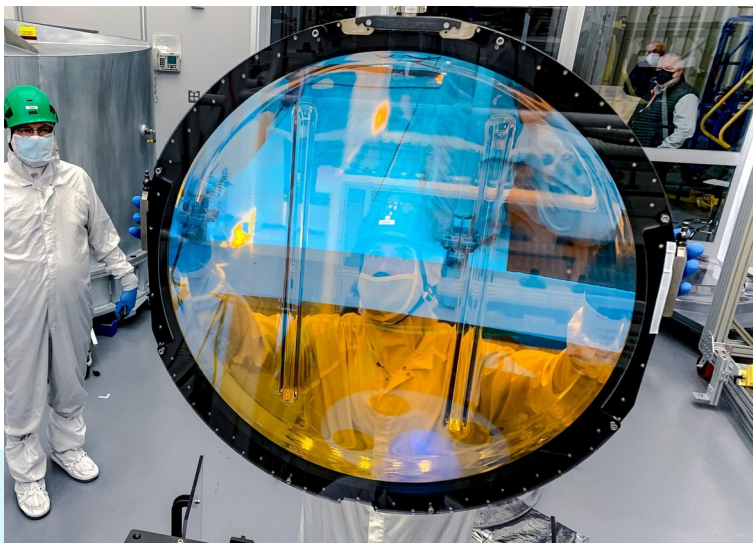
# Narrow & Medium Band & Offset Filters

- Offset: new set of Filters offset by  $\frac{1}{2}$  of the band pass
- Narrow or Medium Band: add multiple narrower filters for more color information
- See HSC Niji project, or J-PAS or PAU



# LSSTCam Filters and Filter Exchange System

- Filters (ugrizy) are 80cm Diameter
- Camera holds 5 Filters at any time, in the Carousel around the cryostat
- Exchange takes 90 seconds + up to 30 seconds to Rotate the Camera



# Narrow Band Filters

- Fast Rubin Beam F# 1.23 limits minimum bandpass achievable
- Beam angle-of-incidence is 14-23 degrees
- From Stubb's slides 2019
- Estimate is that 15-20nm minimum bandpass is achievable

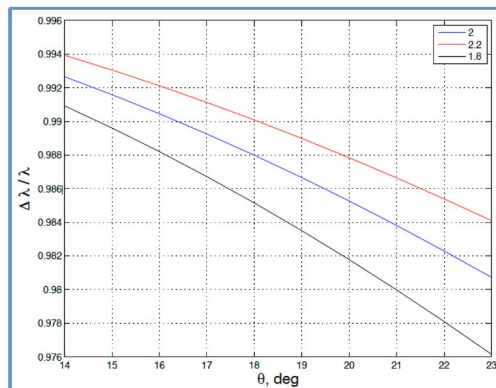
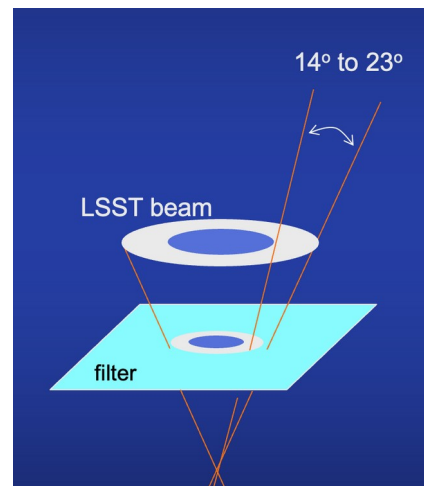
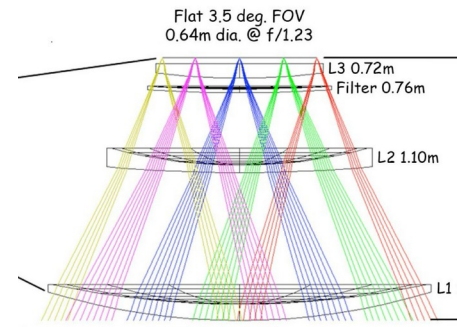


Figure 1. Incidence angle dependence of wavelength shift. The horizontal axis spans the range of angles of incidence in the LSST beam. The vertical axis shows the fractional change in wavelength, compared to  $\theta=0$ , for the effective index values of 1.8, 2.0 and 2.2.



# Narrow Band Filters (cont)

- LSST Filters cost roughly 1M\$ during construction (2014-2021)
- Filters are spherical, so blanks must be figured
- Coating was done by Materion, who built coating chamber to accommodate 80cm filters
- Consider 30 Filters, each 25nm bandpass, to cover entire Si CCD wavelength range
  - At 1M\$ or more each, Quadrant Filters similar to those built for HSC Niji, could be considered
  - Development to coat a single glass blank
  - 4 bandpasses per Filter would reduce cost by a factor
  - Reduce observing inefficiency from Filter exchanges
  - Some loss from mixed throughput from large beam (~12cm diameter) at the Filters



# My Biased Conclusions

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- Augmenting Rubin's Filter set is the most doable, and likely the most interesting, change to Rubin's instrument & capability.
- Other technical directions either require significant R&D, or are prohibitively expensive, for the 2036 time-frame.
- Narrow or Medium Band Filters for F#1.2 require a real design to confirm bandpass and throughput